

Who will leave FreshBasket next quarter?

Loyalty churn prediction and a cheaper way to retain members

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Targeting beats blanket offers

FreshBasket sends retention offers to every member. A churn score lets it spend only where risk is real.

91%

of actual churners reached by
contacting the riskiest 20% of active
members

0.81

PR-AUC for active members on the
held-out label (base rate 9.6%)

1/5

of the blanket-coupon cost for about
the same number of members retained

Most churn is visible early: 80% of churners had already stopped buying before 2024

Behaviour, not demographics: app use, email opens, coupons, purchase momentum and support tickets drive risk

Act: monthly risk list, fix service issues first, coupons for the next tier, prove it with a holdout

Data cleaned before modelling: every fix logged

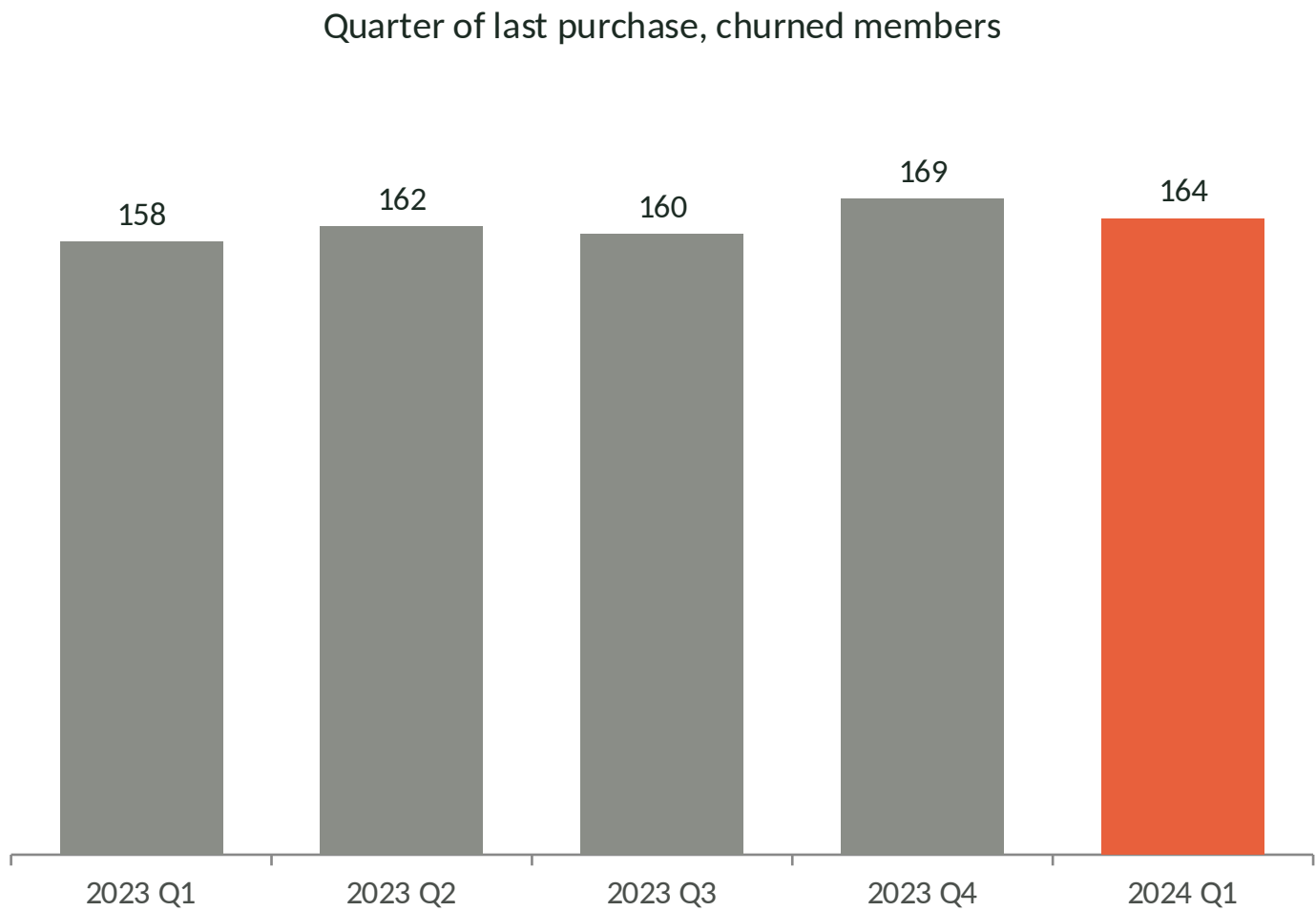
Issue	Found	Action
Duplicates	8 profile, 58 activity, 5 label rows; 2 conflicting customer-months	Dropped
Text casing	104 cities and 104 tiers with stray case/whitespace	Standardised
Spend errors	24 negative, 20 zero with purchases, 13 outliers up to 14.7x	Rebuilt from transactions x basket
Zero-purchase months	1,452 months report categories bought with no purchase	Categories set to 0
Missing months	953 gaps inside 769 member histories	Treated as lost records
Missing values	Age 78, gender 52, home-store price tier 130	Imputed / 'Unknown'

3

leaky label columns removed

LAST_PURCHASE_DATE, MONTHS_OBSERVED and INSUFFICIENT_HISTORY_FLAG are calculated over Apr-Jun 2024. LAST_PURCHASE_DATE alone reproduces the label for every member who ever bought.

Most churners had already gone quiet



80%

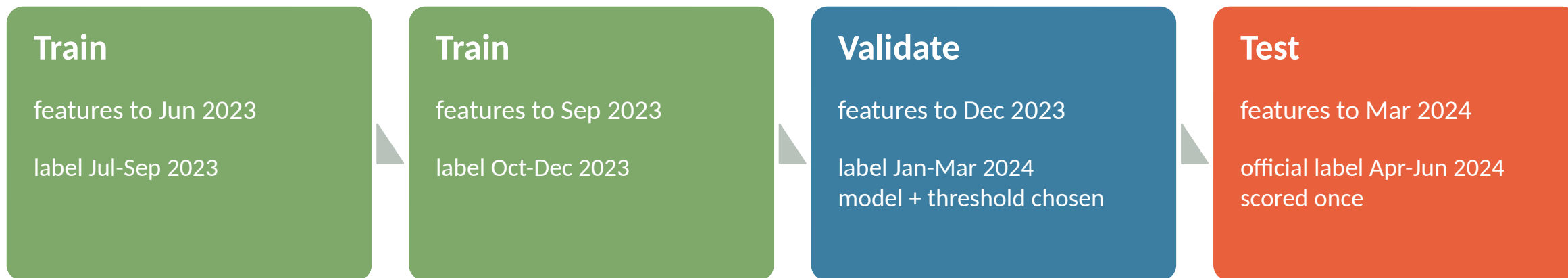
of churners made their last purchase before 2024: recency alone flags them

9.6%

churn among the 1,715 members still buying in Jan-Mar 2024

Every result is reported for both groups. Active members are where a model earns its keep.

Validated on the future, never a random split



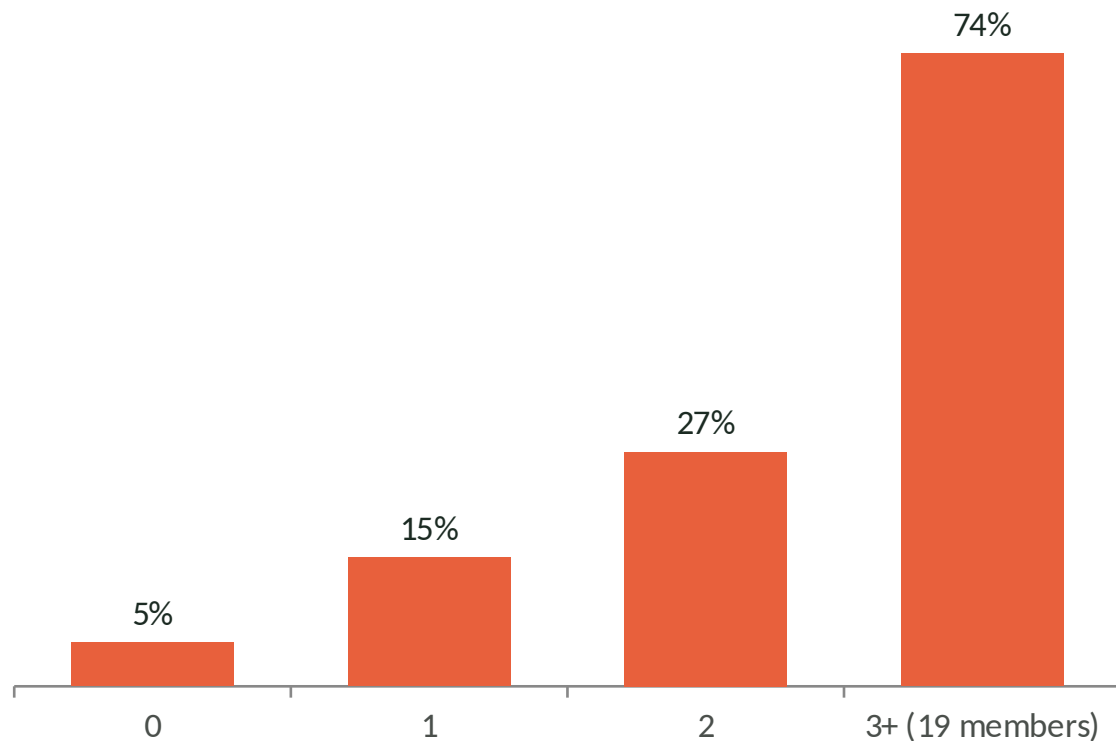
Same churn rule replayed at each cutoff: no purchase in the next 3 months despite earlier purchases

40 leakage-safe features from months up to the cutoff only: recency, purchase and spend trends, rolling app / email / coupon activity, support-ticket trend, tenure, tier and demographics

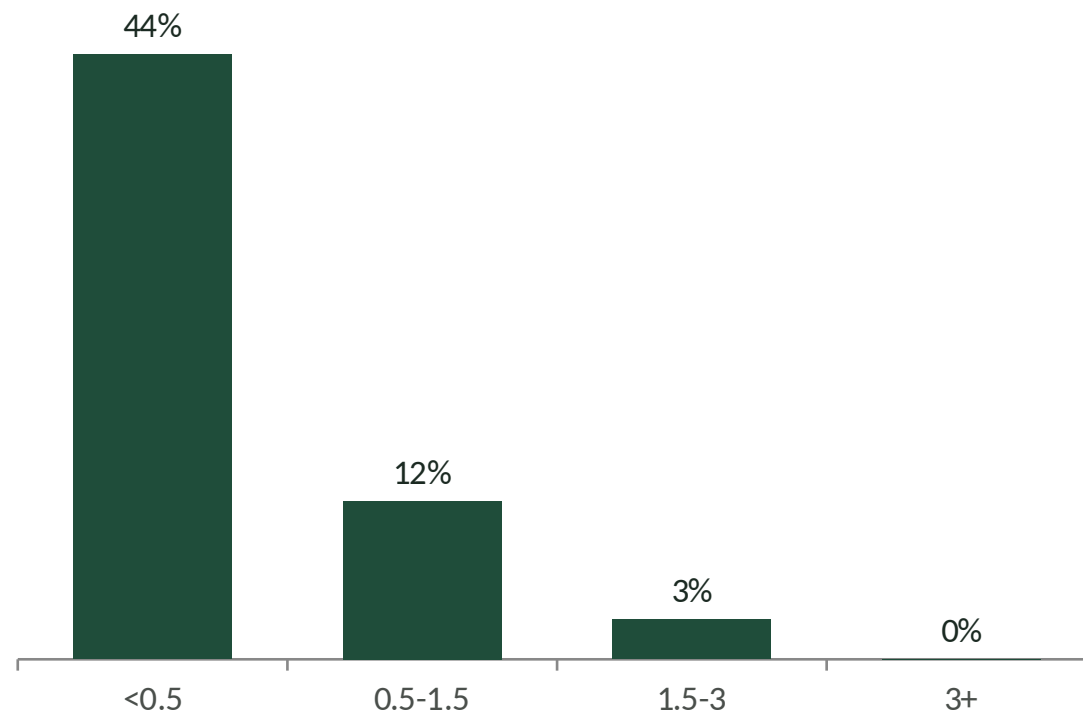
Stable over time: expanding-window backtest gives active-member PR-AUC 0.78-0.82 in every fold

Support friction and disengagement precede churn

Churn rate by support tickets in Jan-Mar

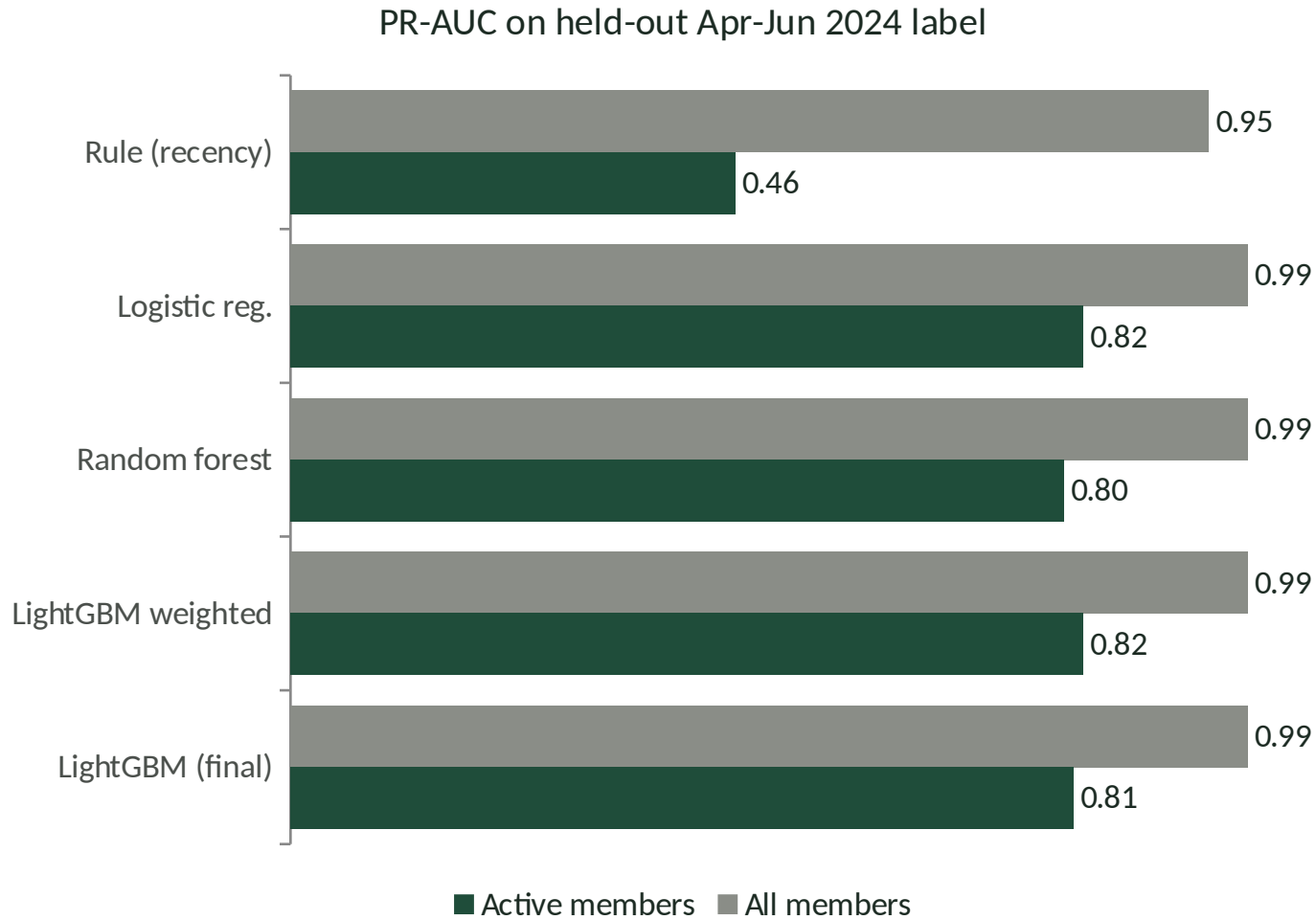


Churn rate by app sessions per month



Active members only. Same pattern for email opens (25% with none vs 2%), coupons (26% vs 2%) and falling purchases. Tier and city barely matter. Associations, not proven causes.

Machine learning beats the recency rule where it matters



0.75 / 0.74

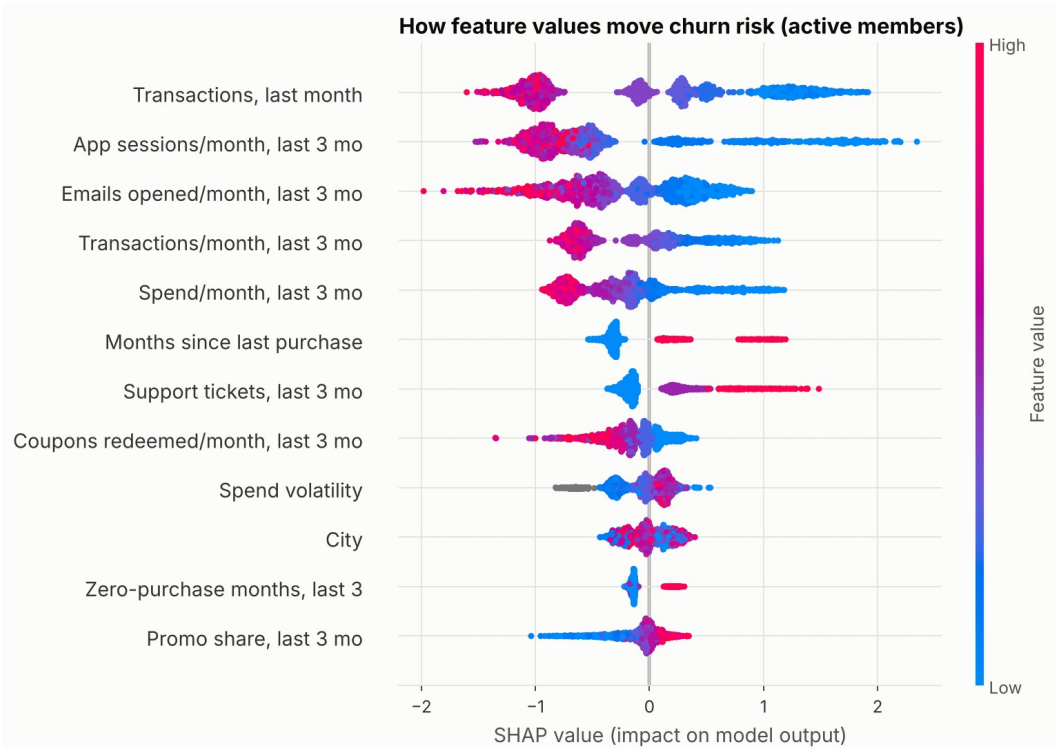
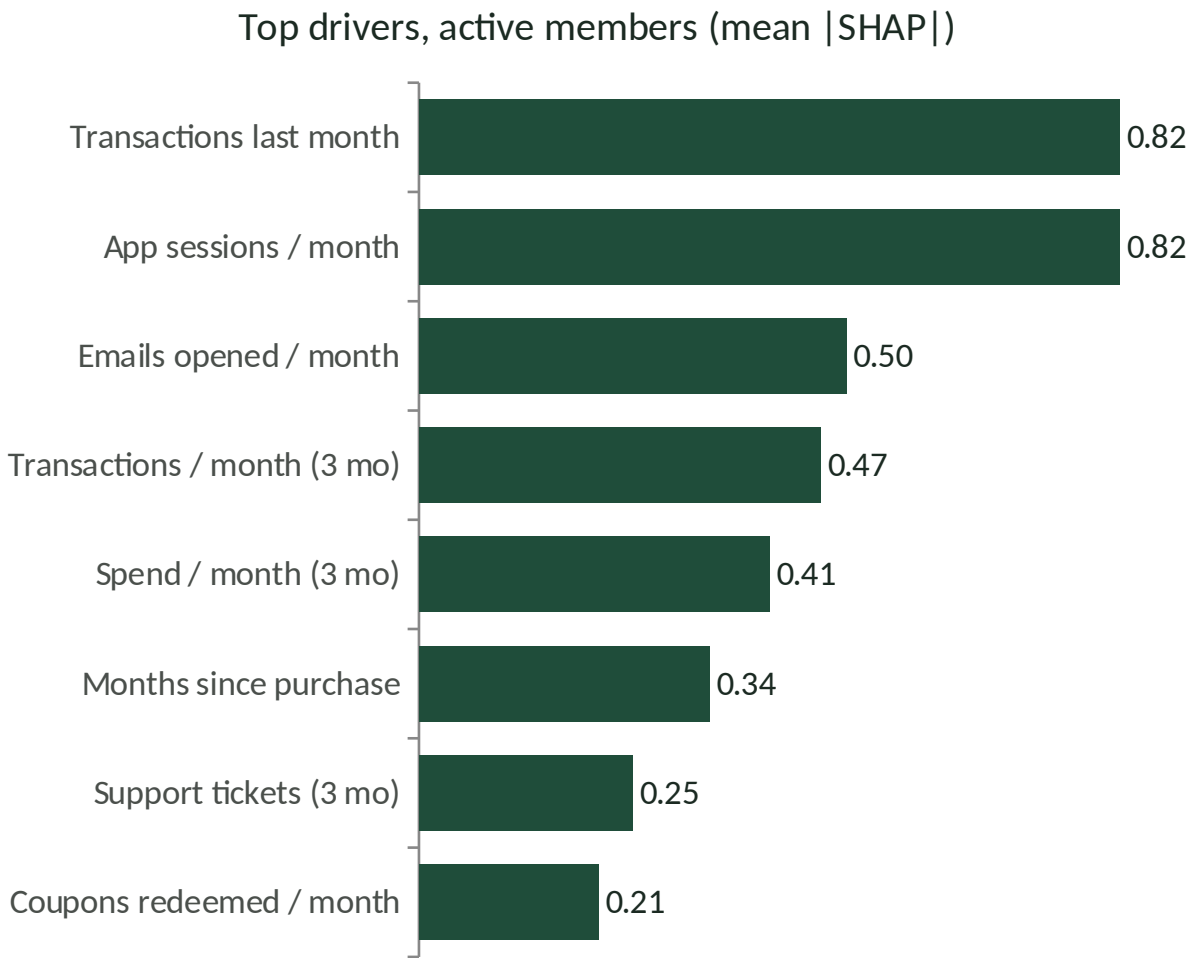
precision / recall for active members at the tuned threshold

Imbalance: threshold tuned for F1 on validation (0.33); class weights changed PR-AUC by under 0.005

Why LightGBM: tied with logistic regression on ranking, but best calibrated (Brier 0.027), which the scenario maths needs

Accuracy alone misleads: "nobody churns" scores 90% on active members

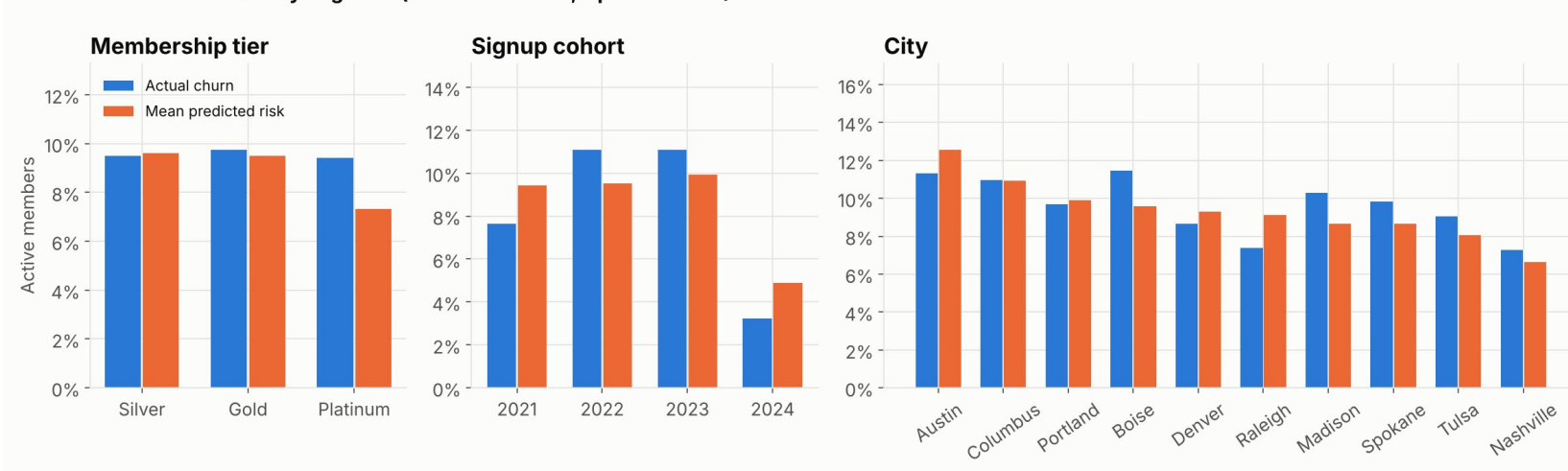
Purchase momentum and app use drive risk



Low purchases, app use, email opens and coupons push risk up (blue dots right); more support tickets push it up (red dots right). Each member's top three drivers are in the prediction file.

Tier and city matter far less than behaviour

Predicted vs actual churn by segment (active members, Apr-Jun 2024)



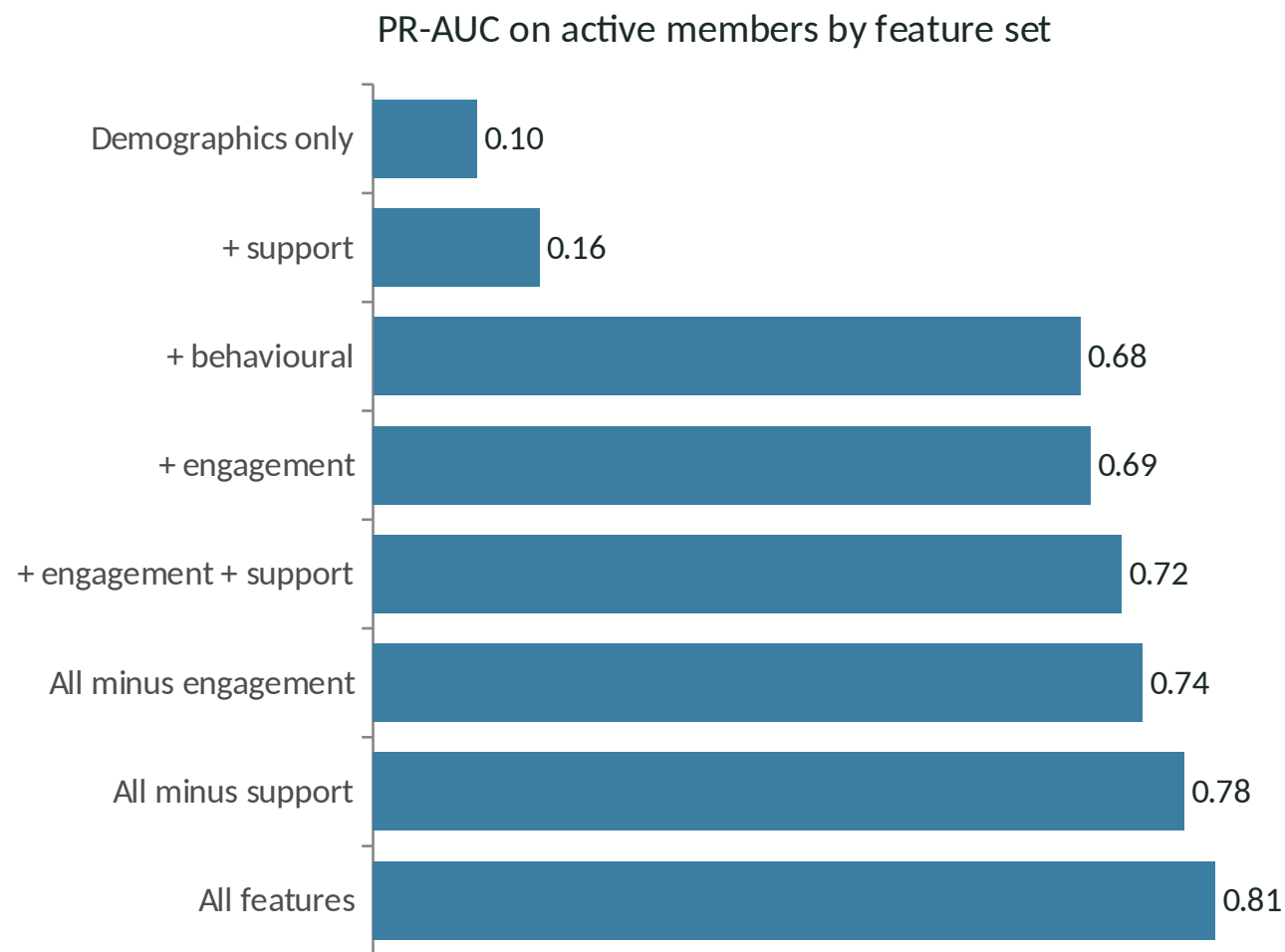
Segment churn among active members ranges only **7-11%**; behaviour moves it **0-74%**

Predictions track actual churn in every tier, cohort and city: no segment is systematically mis-scored

Watch Platinum: actual 9.4% vs predicted 7.3%, a small under-prediction on the most valuable members

2024 joiners churn least: they have had less time to lapse

Engagement and support add what demographics cannot



0.10

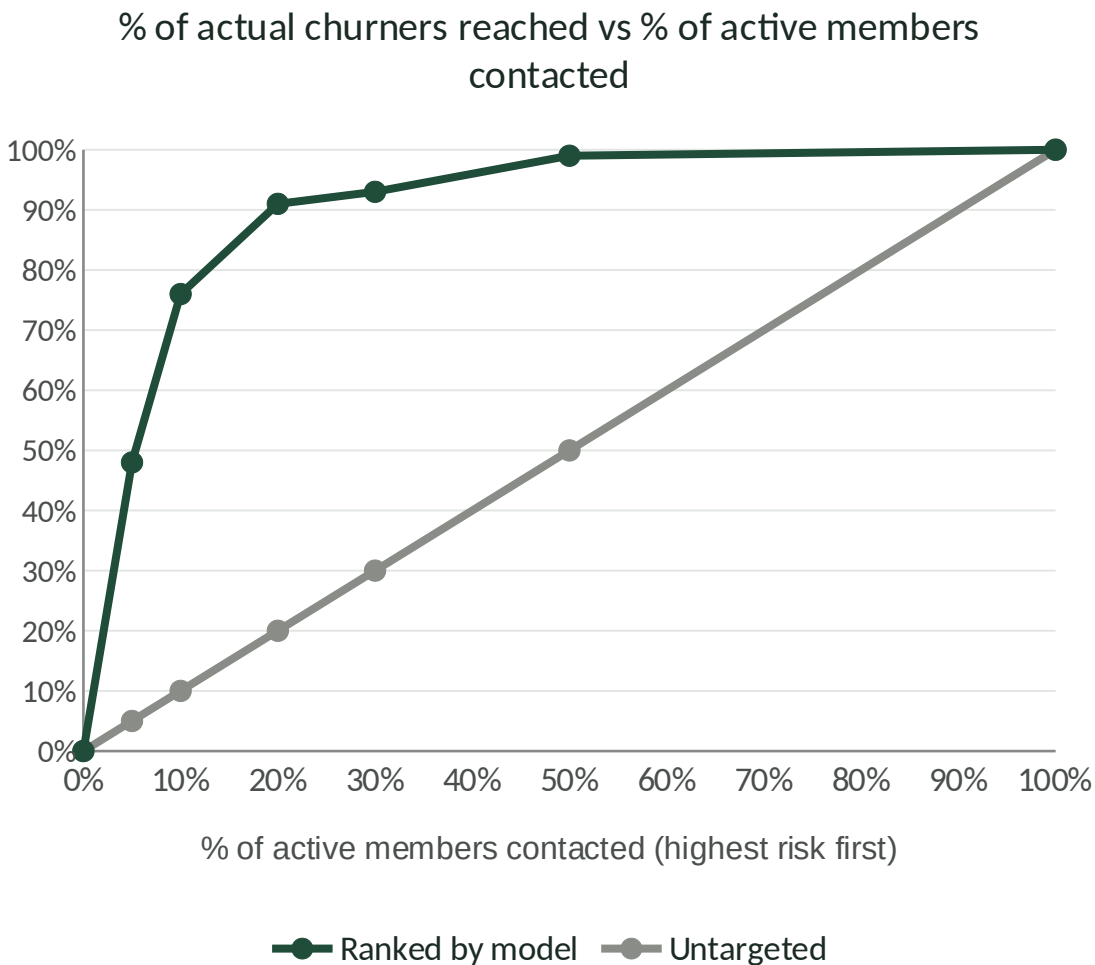
demographics alone: no better than the 9.6% base rate

Engagement is the biggest single lift (0.69)

Support is weak alone but removing it costs 0.03 PR-AUC: it flags members whose purchases still look normal

All groups together: 0.81

A targeted list keeps the savings and cuts the cost



Scenario	Contacted	Retained*	Cost	Cost / retained
A. Service call: ticket + high risk	109	11	\$872	\$77
B. Coupon: top 20% risk	343	20	\$1,715	\$88
C. Coupon: everyone (today)	1,715	20	\$8,575	\$419

*Model-implied risk reduction (about 26%) x 50% assumed effectiveness. \$415 annual margin per retained member; \$5 per coupon, \$8 per call.

Top 20% contacted reaches 91% of churners. Targeted coupons retain about as many as a blanket coupon for one-fifth of the cost; service calls are cheapest per member retained.

Recommendations

1

Score monthly, stop blanket offers

Fund retention for the top 10-20% of active members by risk.

2

Fix service first

High-risk members with tickets or complaints get a proactive call before any discount.

3

Coupons for the disengaged

High-risk, ticket-free members get a personalised coupon or app nudge.

4

Watch app and email as early KPIs

Under one app session a month or no opens for a quarter triggers review.

5

Separate win-back for lapsed

~650 members silent since 2023 need reactivation, not retention.

6

Prove it with a holdout

Keep 10-20% of each target group untreated to measure real uplift.

Assumptions, limitations and next steps

Assumptions and limitations

Official label is the source of truth; 232 members with no prior purchase not scored

Synthetic 18-month data; expect lower accuracy in production

Snapshots share customers, so folds are not fully independent

Scenario effects are correlational what-ifs with an assumed 50% effectiveness

Fixed 3-month horizon counts paused members as churners

Next steps

Run the randomised holdout and replace the assumed effectiveness with measured uplift

Move to uplift modelling: target members most likely to respond, not just most at risk

Monitor monthly: PR-AUC on the latest closed window, calibration, feature drift

Add time-to-churn (survival) modelling to catch fast-rising risk